Practice workbook · Velocity and acceleration by differentiating

Differentiating displacement or velocity with respect to time and substituting a time.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-25

1. Question 3 · part a · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a only.

3 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = t^2 - 4t + 7$$

(a) Find the acceleration of P, in m/s^2, when $t = 3$

(2)

(b) Find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 6$

(4)

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1. Question 3 · part a · 2 marks · Working and answer

Working and answer

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2. Question 1 · part b · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

1 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = 3t^2 - 9t + 7$$

(a) Show that P never comes to rest.

(2)

(b) Find the acceleration of P, in m/s^2, when $t = 4$

(2)

Using algebra

(c) find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 5$

(3)

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2. Question 1 · part b · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 A particle P is moving in a straight line. The displacement s of P, in metres, at time t seconds, $t \geqslant 0$, is given by

$$s = e^{2t} \sin 3t + 2$$

At time $t = 0$, P is at the point A and at time $t = \frac{\pi}{6}$, P is at the point B

(a) Find the exact distance AB

(2)

(b) Find the exact velocity of P when $t = \frac{\pi}{3}$

(4)

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3. Question 6 · part a · 2 marks · Working and answer

Working and answer

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4. Question 6 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 A particle P is moving in a straight line. The displacement s of P, in metres, at time t seconds, $t \geqslant 0$, is given by

$$s = e^{2t} \sin 3t + 2$$

At time $t = 0$, P is at the point A and at time $t = \frac{\pi}{6}$, P is at the point B

(a) Find the exact distance AB

(2)

(b) Find the exact velocity of P when $t = \frac{\pi}{3}$

(4)

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4. Question 6 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 6 continued

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(Total for Question 6 is 6 marks)

4. Question 6 · part b · 4 marks · Working and answer

Working and answer

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5. Question 2 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 A particle P is moving along a straight line, which passes through the fixed point O.

At time t seconds ($t \geqslant 0$), the velocity, v m/s, of P is given by

$$v = t^2 - 3t + 4$$

At time t seconds the acceleration of P is a m/s^2

(a) Find an expression for a in terms of t

(2)

The displacement of P from O is 7 m when $t = 2$

(b) Find the exact displacement of P from O when $t = 4$

(5)

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5. Question 2 · part a · 2 marks · Working and answer

Working and answer

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6. Question 4 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

4 A particle P is moving along the x-axis. At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by $v = 2t^2 - 16t + 30$

(a) Find the acceleration, in m/s^2, of P when $t = 5$

(2)

P comes to instantaneous rest at the points M and N at times t_1 seconds and t_2 seconds where $t_2 > t_1$

(b) Find the exact distance MN

(8)

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6. Question 4 · part a · 2 marks · Working and answer

Working and answer

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7. Question 5 · part a · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 A particle P is moving along the x-axis.

At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

(a) Find the acceleration of P when $t = 5$

(2)

The particle comes to instantaneous rest at the points A and B at times t_1 seconds and t_2 seconds respectively, where $t_1 < t_2$

(b) Find the value of t_1 and the value of t_2

(2)

(c) Use calculus to find the distance AB

(3)

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7. Question 5 · part a · 2 marks · Working and answer

Working and answer

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8. Question 6 · part a · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a only.

6 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the displacement, s metres, of P from the origin O, is given by $s = t^3 - 4t^2 - 16t - 8$

(a) Find the distance of P from O when $t = 0$

(1)

(b) Find the value of t for which P is instantaneously at rest.

(4)

(c) Find the value of t for which P is accelerating at $10\,\text{m/s}^2$ in the positive x direction.

(3)

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8. Question 6 · part a · 1 mark · Working and answer

Working and answer

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9. Question 2 · part a · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

2 A particle P moves along the x-axis. At time t seconds, the displacement, x metres, of P from the origin O is given by

$$x = t^4 - 13.5t + 12$$

(a) Find the velocity, in m/s, of P when $t = 3$

(2)

(b) Find the value of t for which P is instantaneously at rest.

(2)

(c) Find the acceleration, in m/s^2, of P when $t = 2$

(2)

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9. Question 2 · part a · 2 marks · Working and answer

Working and answer

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10. Question 2 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

2 A particle P moves along the x-axis. At time t seconds, the displacement, x metres, of P from the origin O is given by

$$x = t^4 - 13.5t + 12$$

(a) Find the velocity, in m/s, of P when $t = 3$

(2)

(b) Find the value of t for which P is instantaneously at rest.

(2)

(c) Find the acceleration, in m/s^2, of P when $t = 2$

(2)

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10. Question 2 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 2 continued

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10. Question 2 · part c · 2 marks · Working and answer

Working and answer

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11. Question 7 · part a · 1 mark

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

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11. Question 7 · part a · 1 mark · Working and answer

Working and answer

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12. Question 7 · part c · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

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12. Question 7 · part c · 2 marks · Working and answer

Working and answer

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